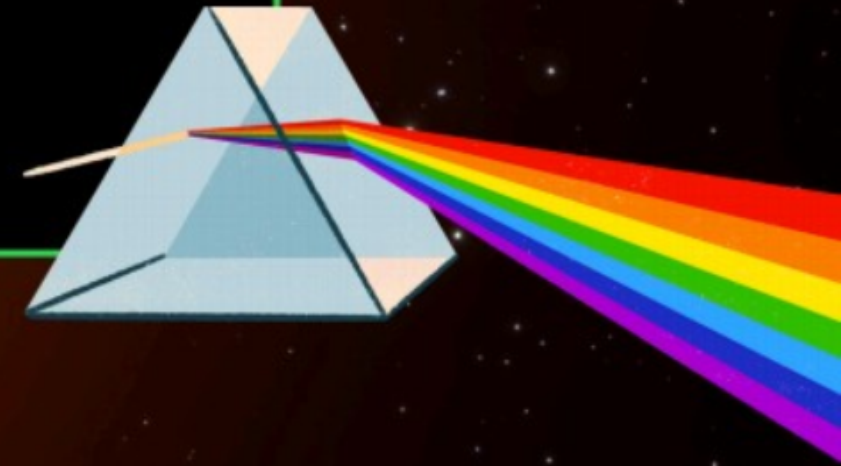
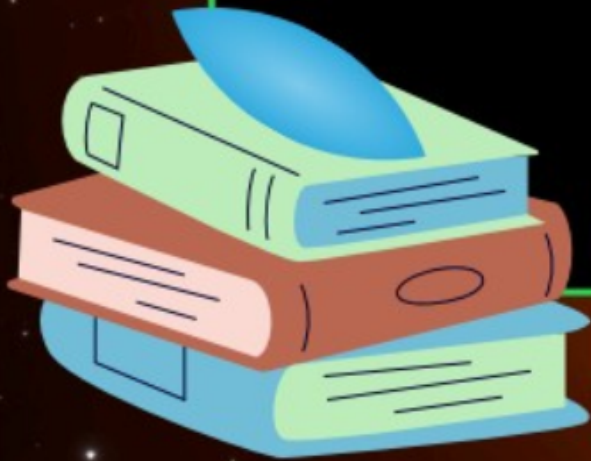


CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD



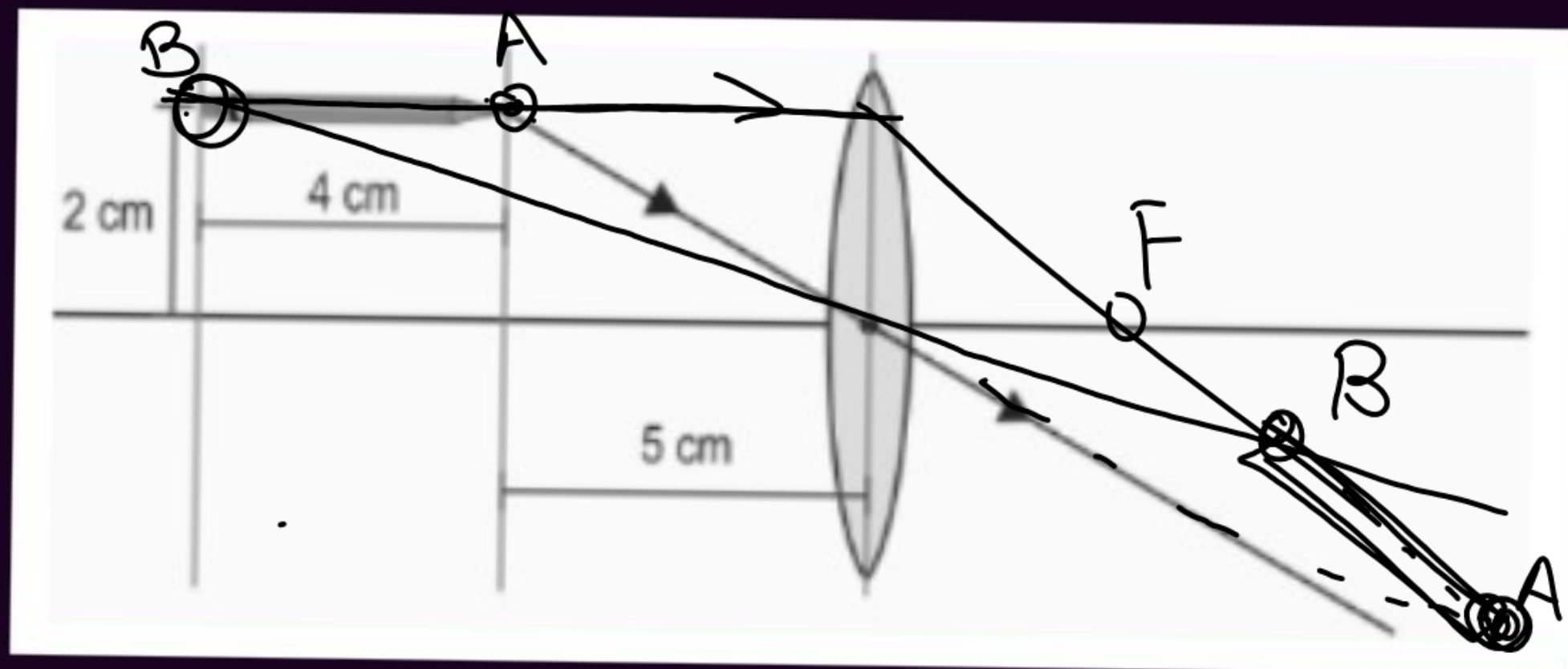
Point A

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m.

f the whole

(lens.)

end of the

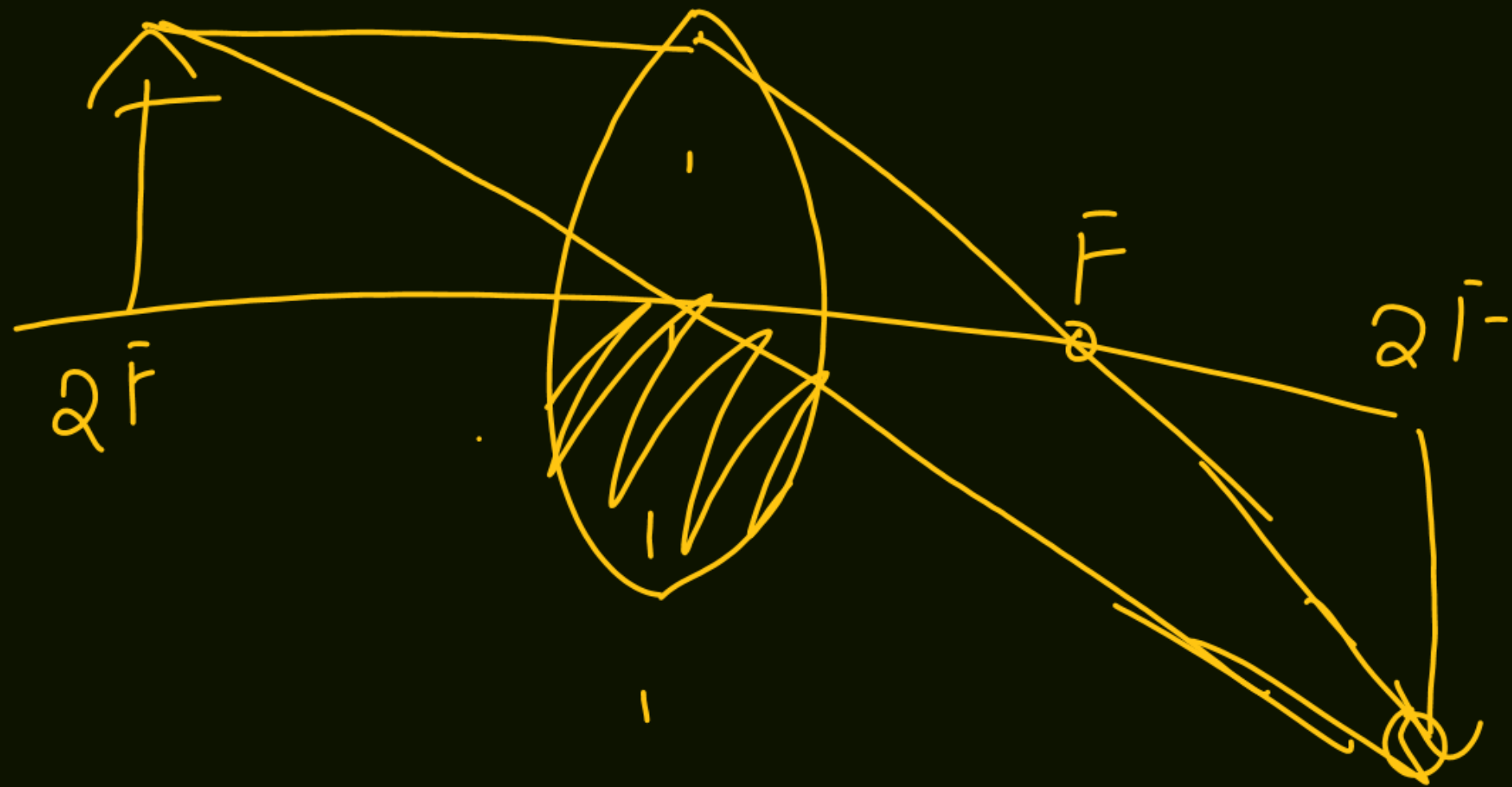


Soch ke bataao

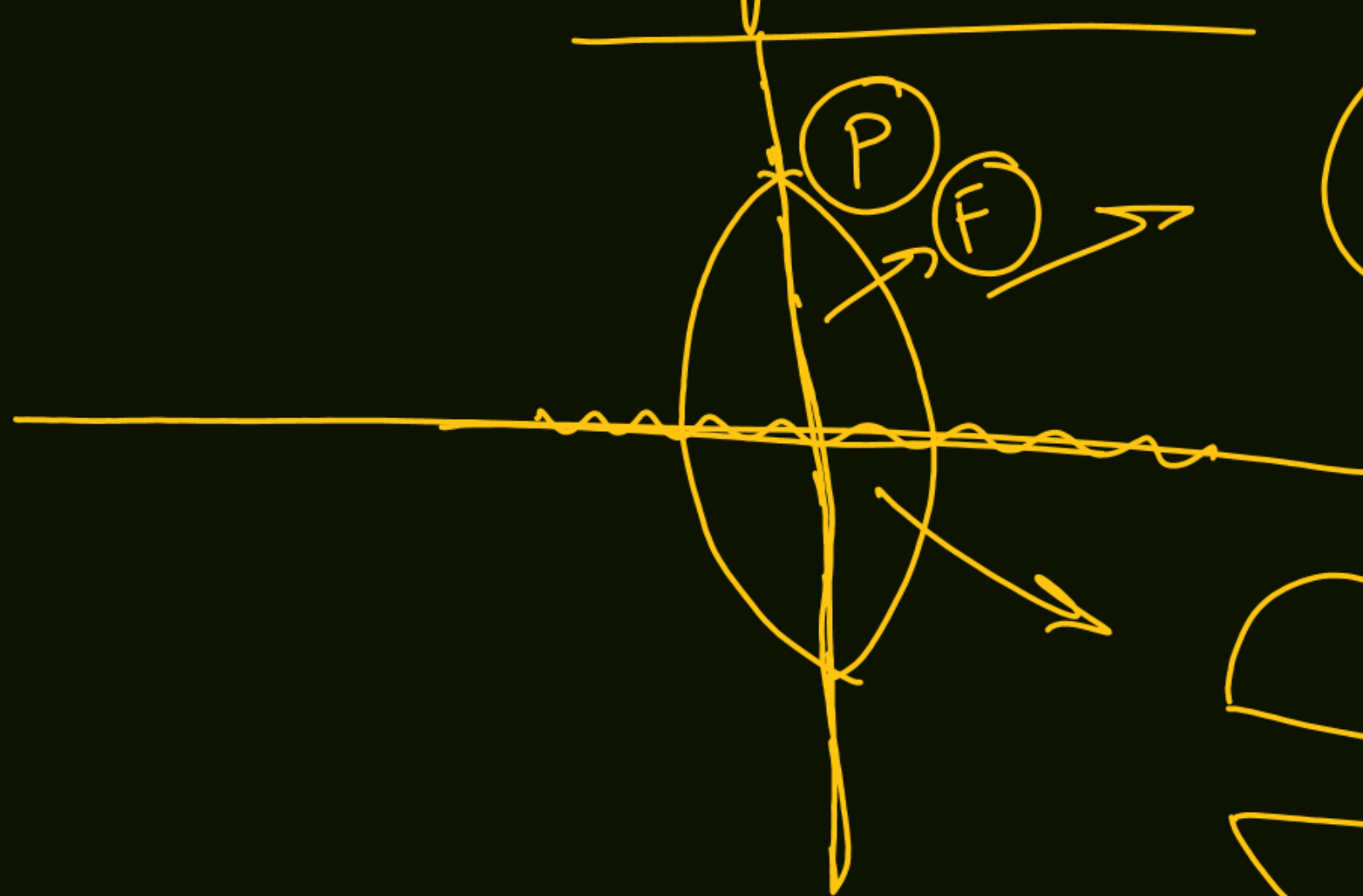
- Q. (a) The upper half of a convex lens is covered with black paper. Draw a ray diagram to show the formation of image of an object placed at a distance of $2F$ from such a lens. Mention the position and nature of the image formed. State the observable difference in the image obtained if the lens is uncovered. Give reason to justify your answer.
- (b) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 15 cm. Use lens formula to determine the distance of the image from the optical centre of the lens.

HW

Kya
ye
Sahi
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④ Im concept



$$\boxed{P' = \frac{P}{2}} \rightarrow F' = 2F$$



LPSS
Bridgford

$$\begin{aligned} P' &= P \\ F' &= F \end{aligned}$$

$$\begin{aligned} P' &= P \\ F' &= F \end{aligned}$$

Soch ke bataao

Q. Smriti is looking at herself in a convex mirror, in a science museum, standing 2 m away from the mirror. Her image appears to be around half her actual height. Estimate the focal length of the mirror.

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$u = -2m$$

$$f = ?$$



convex

Virtual

$$m = \frac{1}{2}$$

$$m = -\frac{v}{u} = \frac{1}{2}$$

$$v = -\frac{u}{2}$$

$$v = ? \quad 1m$$

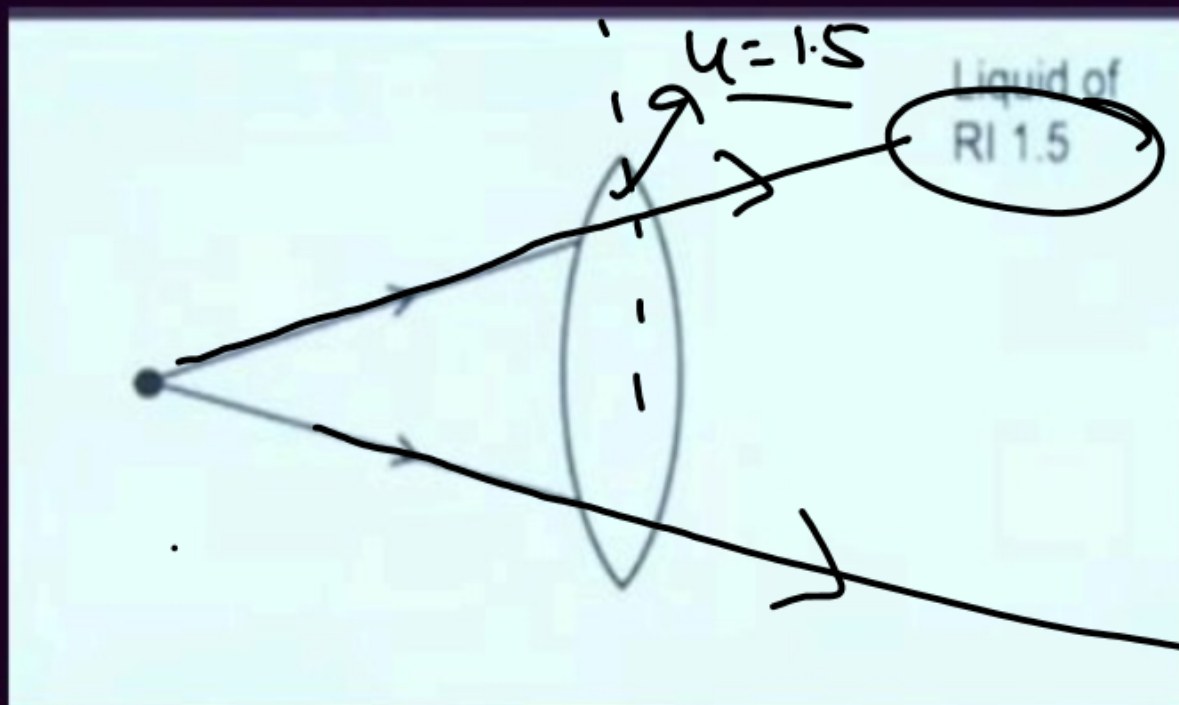
Soch ke bataao

Q. A convex lens is used to form a real image of an object. State how the image and focal length will be affected in the following cases:

- (i) When half of the lens is covered with black paper. ✓
- (ii) When the lens is cut into two equal halves perpendicular to the principal axis. ✓
- (iii) When the lens is cut into two equal halves parallel to the principal axis. ✓

Soch ke bataao

Q. A lens made of a material with refractive index 1.5 is immersed in a liquid with refractive index 1.5. The diagram shows two rays incident on the lens when it is immersed in the liquid. Complete the diagram and draw the light rays after they pass through the lens. Justify your diagram.



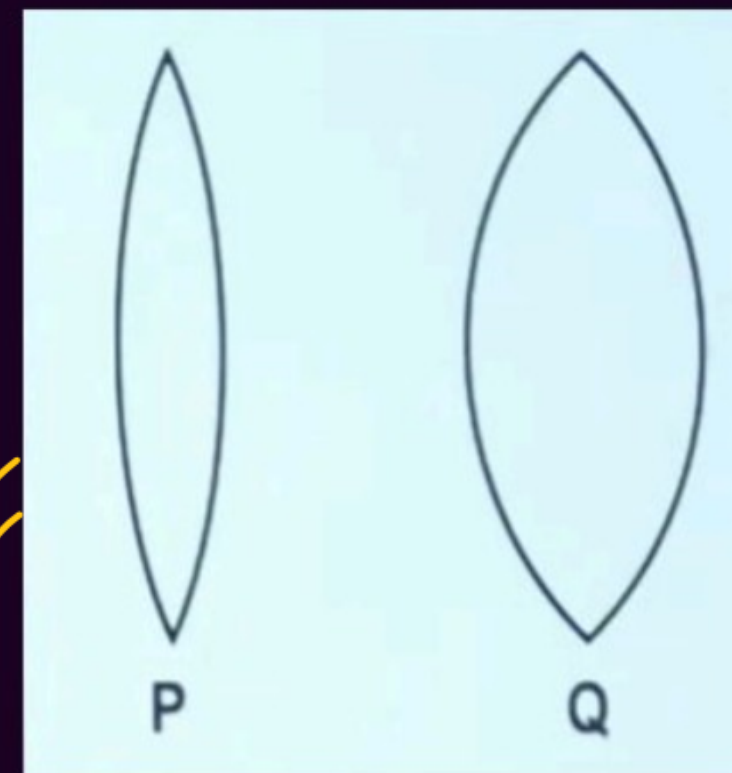
Soch ke bataao

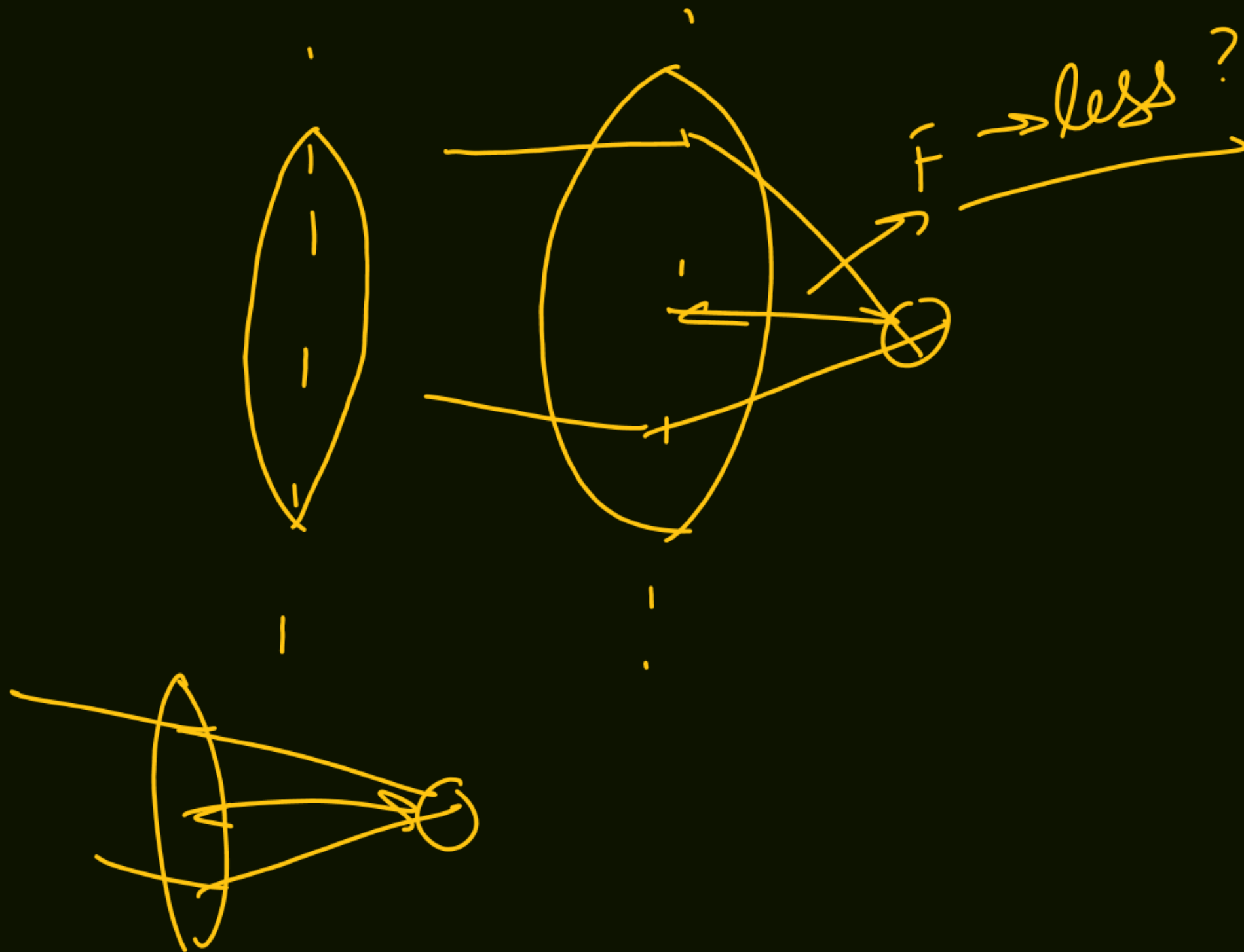
Manju

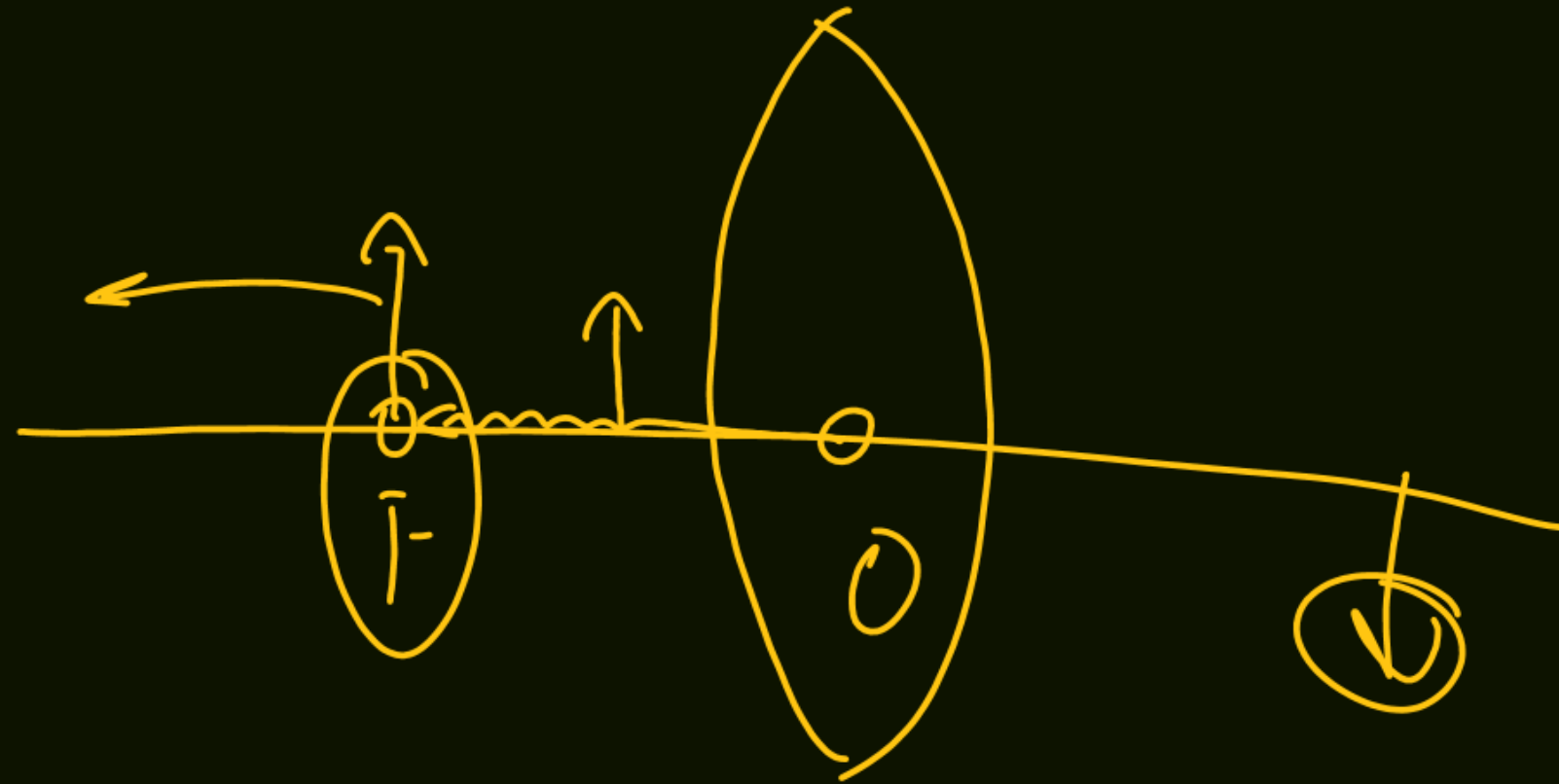
Q. Manju has two convex lenses P and Q made of the same material as shown in the figure.

She is looking at printed text on a page using lens P. She notices that when she slowly moves the lens away from the page, the text turns upside down when the lens crosses a distance of 10 cm from the page.

- (a) Which characteristic of lens P does the distance 10 cm signify?
- (b) If Manju does the same experiment with lens Q, at what distance will the inversion of the image happen – less than, equal to, or more than 10 cm? Justify your answer.







Soch ke bataao

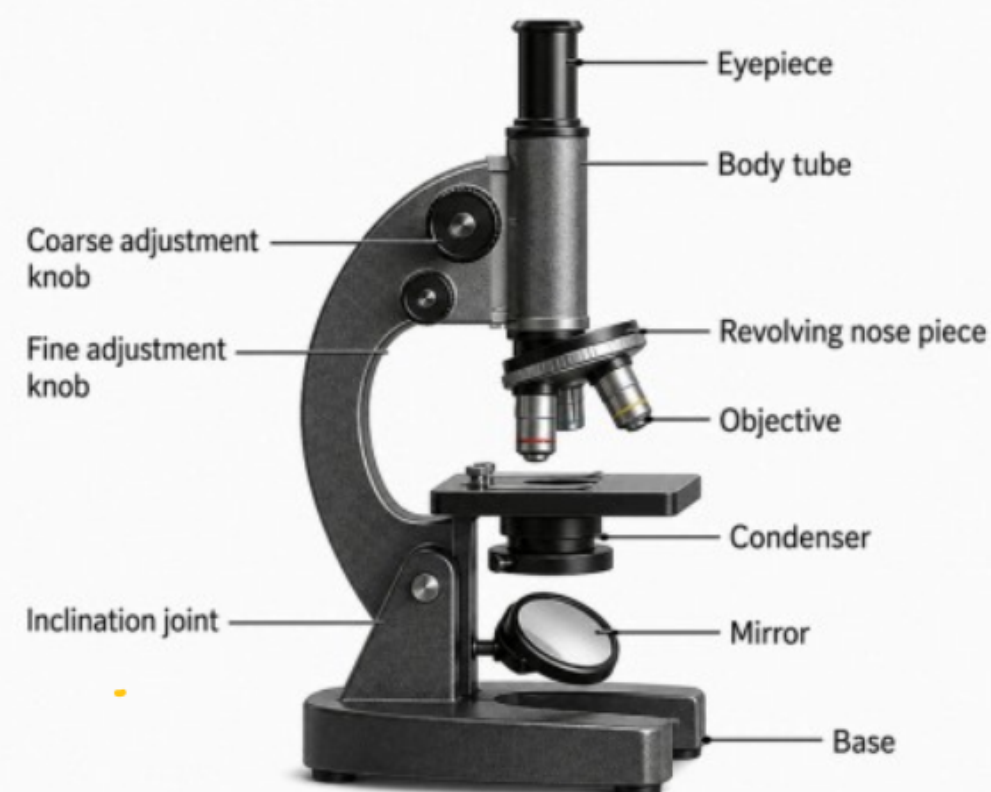
Q. Read the source below and answer the questions that follow:

A compound microscope is an instrument which consists of two lenses L_1 and L_2 . The lens L_1 , called the objective, forms a real, inverted and magnified image of the given object. This serves as the object for the second lens L_2 , the eyepiece. The eyepiece functions like a simple microscope or magnifier. It produces the final image, which is inverted with respect to the original object, enlarged and virtual.

Q. What types of lenses must be L_1 and L_2 ?

- A. Both concave
- ☒ B. Both convex
- C. L_1 -concave and L_2 -convex
- D. L_1 -convex and L_2 -concave

$L_1 \rightarrow$ convex
 $L_2 \rightarrow$ convex ✓



What is the value and sign of magnification (according to the New Cartesian Sign Convention) of the image formed by L_1 ?

- A. Value less than 1 and sign positive
- B. Value more than 1 and sign positive
- C. Value less than 1 and sign negative
- D. Value more than 1 and sign negative

→ Real
 $m = -1$ ↑

What is the value and sign of magnification (according to New Cartesian Sign Convention) of the image formed by L_2 ?

- A. Value less than 1 and sign positive
- B. Value more than 1 and sign positive
- C. Value less than 1 and sign negative
- D. Value more than 1 and sign negative

$m = +2$ ↑
 $P = 5D = \frac{1}{f}$ ✓

If power of the eyepiece (L_2) is 5 diopters and it forms an image at a distance of 80 cm from its optical centre, at what distance should the object be?

- A. 12 cm
- B. 16 cm
- C. 18 cm
- D. 20 cm

$v = 80\text{cm}$ ✓

